

SSDI Project - Calorease User Manual

Team-5

1. Deployment and Installation Steps

- Technical Specifications:
 - Operating System: Windows 10/11, macOS 13+, or Linux (Ubuntu 20.04+).
 - Dependencies:
 - Frontend: React.js, Node.js (v14.0+).
 - Backend: Express.js, MySQL (for database management).
 - Others: NPM (v6.0+), MySQL Workbench or Command-Line Client for database setup.
- Steps to Deploy/Install:
 - a. Before starting, ensure you have Node.js and npm installed on your machine. If not, download and install them from nodejs.org and git installed from github.com/downloads.
 - b. Clone the Repository:
 - Open a terminal and run the following:
 - `git clone https://github.com/naveenaduri/calorease.git`
 - `cd calorease`
 - c. Install Backend Dependencies:
 - Navigate to the backend folder and run:
 - `cd backend`
 - `npm install`
 - d. Install Frontend Dependencies:
 - Navigate to the frontend folder and run:
 - `cd ../frontend`
 - `npm install`
 - e. Set Up the MySQL Database:
 - Open MySQL Command Line Client and log in using your credentials.
 - Create Data using below command in mysql:
 - `Create database final_project_ssdi;`
 - Confirm the database is created by running:
 - `SHOW DATABASES;`
 - Use the below commands to Create and Grant Permissions to the user

- `CREATE USER 'ssdi_backend_user'@'host' IDENTIFIED BY 'Qwerty567@';`
- `GRANT ALL PRIVILEGES ON final_project_ssdi.* to 'ssdi_backend_user'@'host';`
- Replace host in above two commands with your host variable.

- Import the MySQL dump file:
 - `cd /path-to-calorease-repository`
 - `SOURCE /final_project_ssdi.sql;`

f. Start the Servers:

- Navigate to the backend project and run:
 - `node server.js`
- Navigate to the frontend project and run:
 - `npm run start`

g. Access the Application:

- Open your browser and go to:
 - <http://localhost:3000>

2. Main Features of Calorease

Calorease offers a set of powerful tools to help users track their fitness and diet efficiently. Here are the key features:

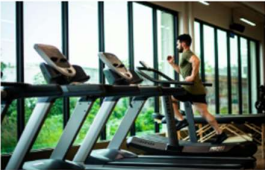
CalorEase

HOME SIGNUP LOGIN



USER PAGE

Enables user registration with personal details, email, password, and optional verification for security.



WORKOUT TRACKER

Records the number of calories burned and total minutes spent exercising for each workout session.



DIETARY INFORMATION

Allows users to log daily meals and monitor caloric intake to align with their fitness goals.



CALORIES BURNED SUMMARY

Displays an overview of total calories burned over time, helping users stay motivated and track progress.



WORKOUT ANALYSIS

Provides insights into total workout minutes each week, visualizing commitment to fitness goals.



PROGRESS INSIGHTS

Combines calorie burn and workout duration trends to show improvements in fitness over time.

- **User Registration**

- Signup Page: New users can sign up by filling out a form with the following details:
 - First Name
 - Last Name
 - Email
 - Phone Number
 - Password & Confirmed Password
 - Date of Birth
 - Weight
 - Height
- The system ensures that all fields are correctly filled.
 - Passwords must match the confirmation field. If not, an error will be displayed.
 - Basic field validations such as valid email format and password strength are implemented.

CalorEase

HOME SIGNUP LOGIN

First Name

Team

Last Name

3,14

Email

Team314@gmail.com

Phone Number

1234567890

Password

Confirm Password

Gender

Male

Birth Date

21-11-2024

Height (cms)

155

Initial Weight (lbs)

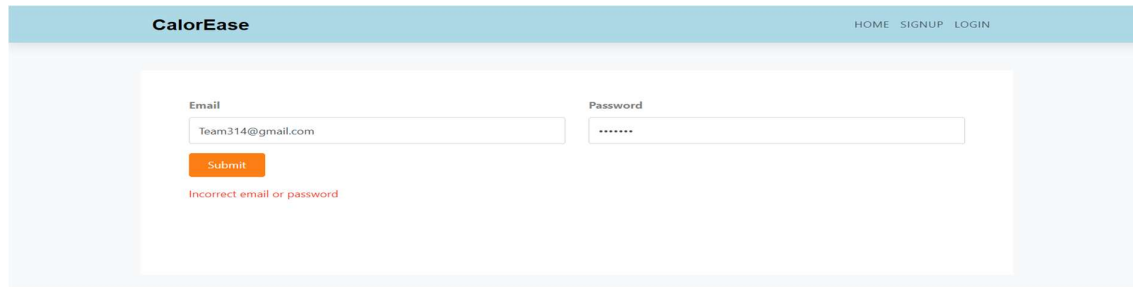
140

Submit

Registration successful!

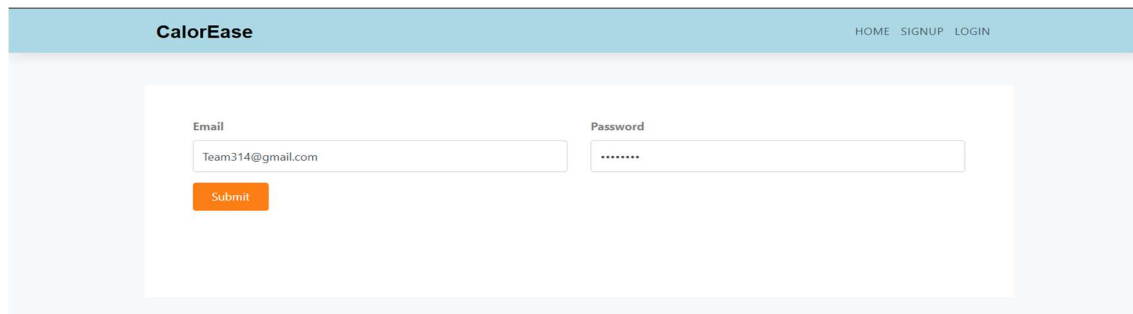
- **Login**

- Once the account is created, the user can log in by entering:
 - **Email Address** (same as the one used during signup)
 - **Password**
- If the user enters incorrect credentials, an **error message** will be displayed, indicating that the password or email is incorrect.



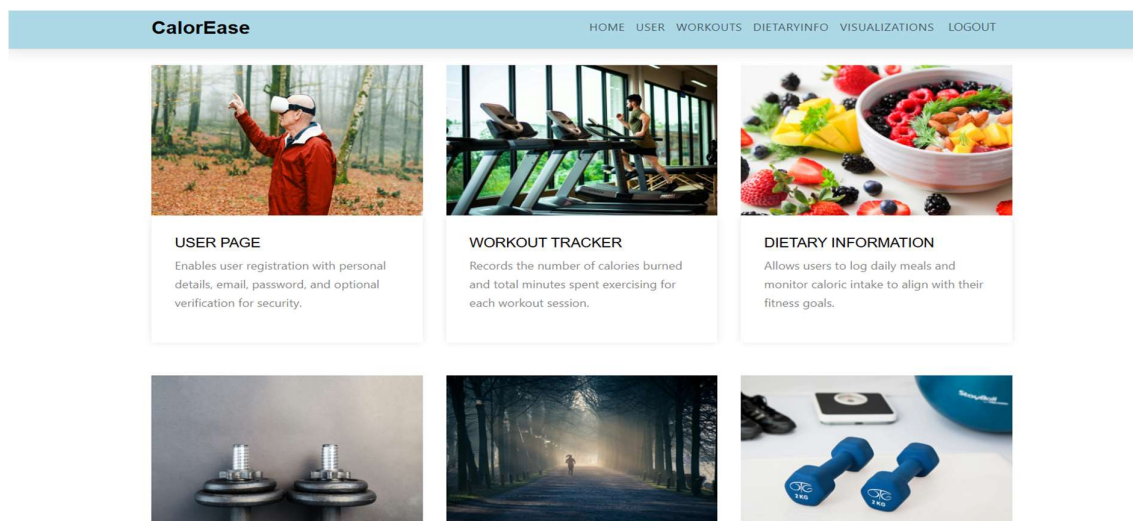
The screenshot shows the CalorEase login interface. At the top, there is a navigation bar with the logo 'CalorEase' on the left and links for 'HOME', 'SIGNUP', and 'LOGIN' on the right. Below the navigation bar is a white login form. The form has two input fields: 'Email' and 'Password'. The 'Email' field contains the text 'Team314@gmail.com'. The 'Password' field contains a series of dots. Below the 'Email' field is an orange 'Submit' button. Below the 'Submit' button, there is a red error message that reads 'Incorrect email or password'.

- On successful login, the user will be redirected to the **Home Page** which are mentioned below.



This screenshot is identical to the one above, showing the CalorEase login interface. However, in this version, the 'Submit' button is highlighted in orange, indicating a successful login. The error message is no longer present.

- After logging in, users are directed to the home page, where they can navigate to the following sections:
 - **User Page:** View and manage their personal information.
 - **Workout Page:** Track and log workout details such as type, duration, and date.
 - **Dietary Page:** Log meals, track calories, and monitor their dietary intake.



The screenshot shows the CalorEase home page. At the top, there is a navigation bar with the logo 'CalorEase' on the left and links for 'HOME', 'USER', 'WORKOUTS', 'DIETARYINFO', 'VISUALIZATIONS', and 'LOGOUT' on the right. Below the navigation bar is a grid of six cards, each representing a different section of the application. Each card has a header image, a title, and a brief description of its functionality.

Section	Description
USER PAGE	Enables user registration with personal details, email, password, and optional verification for security.
WORKOUT TRACKER	Records the number of calories burned and total minutes spent exercising for each workout session.
DIETARY INFORMATION	Allows users to log daily meals and monitor caloric intake to align with their fitness goals.
WORKOUT TRACKER (Visuals)	Images of a person running on a treadmill and a person walking on a path.
DIETARY INFORMATION (Visuals)	Images of a bowl of fruit and two blue dumbbells.

- **User Page**

On the User Page (displayed only after successful login), users can manage their personal information. They can edit fields such as **First Name**, **Last Name**, **Date of Birth**, and **Weight**. The system does not allow users to change their email address.

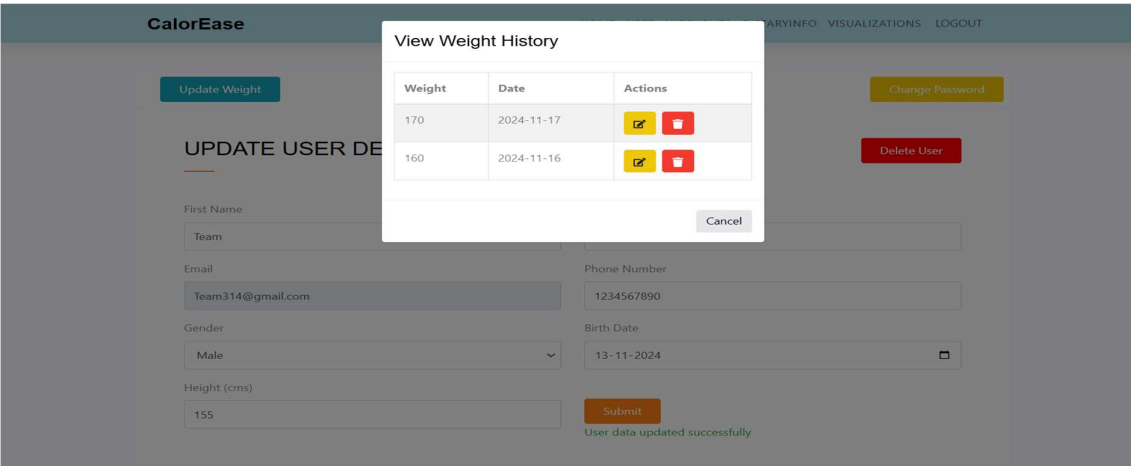
The screenshot shows the 'CalorEase' user interface. At the top, a navigation bar includes 'HOME', 'USER', 'WORKOUTS', 'DIETARYINFO', 'VISUALIZATIONS', and 'LOGOUT'. Below this, a header row contains three buttons: 'Update Weight' (blue), 'View Weights' (dark grey), and 'Change Password' (yellow). The main content area is titled 'UPDATE USER DETAILS' and features a 'Delete User' button (red) in the top right corner. The form contains several input fields: 'First Name' (text box with 'Team'), 'Last Name' (text box with '3.14'), 'Email' (text box with 'Team314@gmail.com'), 'Phone Number' (text box with '1234567890'), 'Gender' (dropdown menu with 'Male' selected), 'Birth Date' (calendar icon with '13-11-2024'), and 'Height (cms)' (text box with '155'). A 'Submit' button (orange) is located at the bottom right of the form. Below the form, a green message states 'User data updated successfully'.

There are separate buttons for:

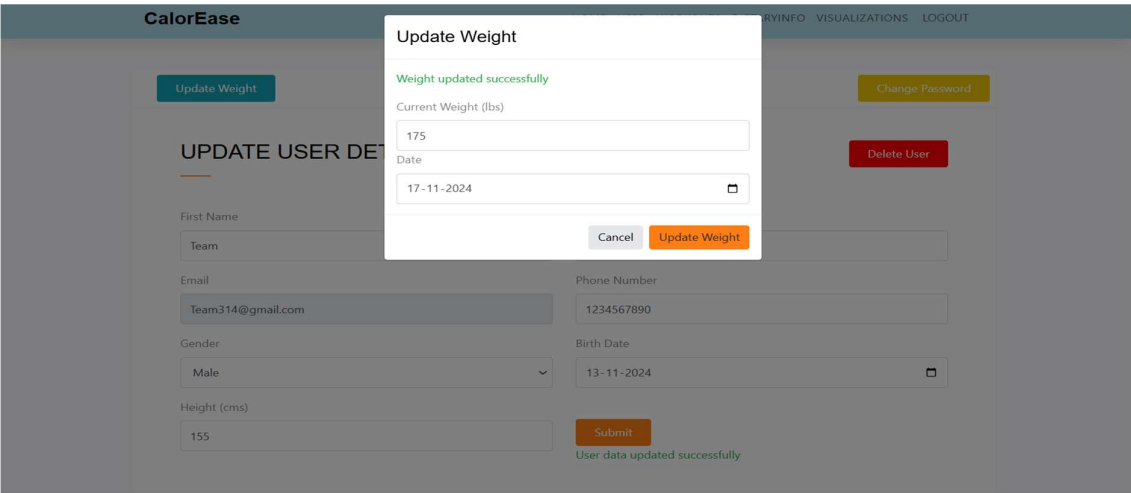
- **Deleting Account:** Users can delete their account if they wish to remove their data from the system.
- **Changing Password:** Users can change their password for better security.
- **Updating Weight:** Users can update their weight, and they can also view their previous weights by clicking the **View Weight** button.

This screenshot shows the same 'CalorEase' user interface as the previous one, but with the 'Update Weight' modal open. The modal is a white box with a title 'Update Weight' and a green message 'Weight updated successfully'. It contains two input fields: 'Current Weight (lbs)' (text box with '170') and 'Date' (calendar icon with '17-11-2024'). At the bottom of the modal are two buttons: 'Cancel' (grey) and 'Update Weight' (orange). The background shows the 'UPDATE USER DETAILS' form and the 'Delete User' button, which are slightly dimmed. The 'Submit' button and the 'User data updated successfully' message are also visible in the background.

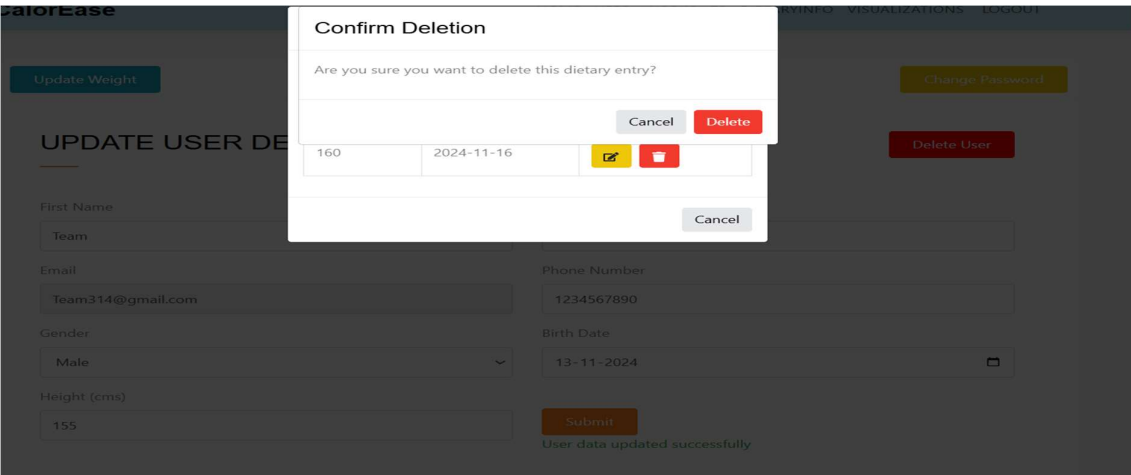
- **View Weight :**The **View Weight** section displays a table of previously logged weight entries. This table includes icons that allow users to:



- **Edit Weight:** Users can edit the weight entry if they want to update any details.



- **Delete Weight:** Users can delete a specific weight entry from the system if it is no longer needed



- **Workout Page**

On the Workout Page, users can track and log their workout details. The following information can be entered for each workout:

CalorEase

HOMEUSERWORKOUTSDIETARYINFOVISUALIZATIONSLOGOUT

Workout Information

Add New Workout

Select Date

dd-mm-yyyy

Workout	Description	Duration	Calories Burned	Date	Actions
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- **Workout Name:** The name of the exercise performed (e.g., Running, Yoga).
- **Duration:** The length of time the workout was performed.
- **Calories Burnt:** The estimated calories burned during the workout.
- **Date:** The date when the workout took place.
- **Description:** A brief description of the workout, if needed (e.g., type of training or focus area).
- Upon selecting a specific date on the page, users can view their logged workouts for that date in a table format. The table displays the workout details entered, making it easy to track progress.

Users can add a new workout by clicking the Add Workout button. This opens a form where they can input the workout information.

CalorEase

DIETARYINFOVISUALIZATIONSLOGOUT

Workout Information

Add New Workout

Select Date

16-11-2024

Workout	Description
chest flies	Done with support

Add Workout Entry

Workout Name

chest pullups

Duration (minutes)

10

Calories Burnt

100

Performed On

16-11-2024

Description

Done with support

Add Entry

Close

CalorEase

HOMEUSERWORKOUTSDIETARYINFOVISUALIZATIONSLOGOUT

Workout Information

Add New Workout

Select Date

16-11-2024

Workout	Description	Duration	Calories Burned	Date	Actions
chest flies	Done with support	10	100	2024-11-16	
chest pullups	Done with support	10	100	2024-11-16	

- The table also includes Edit and Delete buttons:
 - Edit Button: Users can edit the details of a workout by clicking on the Edit icon, allowing them to update information such as the name, duration, calories, and description. Any changes made will be immediately reflected in the table.

CalorEase

DIETARYINFOVISUALIZATIONSLOGOUT

Workout Information

Add New Workout

Select Date

16-11-2024

Workout	Description
chest flies	Done with support
chest pullups	Done with support

Edit Workout Entry

Workout Name

chest flies

Duration (minutes)

10

Calories Burnt

100

Performed On

16-11-2024

Description

Done with support

Update Entry

Close

- Delete Button: Users can delete a specific workout entry by clicking the Delete icon. The entry will be removed from the table, ensuring that only relevant data is kept.

- **Dietary Info Page:**

On the Dietary Info Page, users can log their dietary intake, tracking meals and nutritional information. The following details can be entered for each meal:

- Meal Name: The name of the food or meal (e.g., Breakfast, Lunch, Snack).
- Calories: The total calories for the meal.
- Protein: The amount of protein in the meal (in grams).
- Serving Quantity: The quantity or portion size of the meal (e.g., 1 serving, 2 servings).
- Intake Date: The date when the meal was consumed.

CalorEase

HOMEUSERWORKOUTSDIETARYINFOVISUALIZATIONSLOGOUT

Dietary Information

Add New Entry

dd-mm-yyyy

Food Name	Calories	Protein	Serving Quantity	Intake Date	Actions
No entries found					

Upon selecting a specific date, users can view their logged meals for that date in a table format. The table displays the meal details entered, allowing users to monitor their dietary intake.

Users can add a new meal by clicking the Add Meal button. This opens a form where they can input the dietary details for that meal.

CalorEase

DIETARYINFOVISUALIZATIONSLOGOUT

Dietary Information

17-11-2024

Add New Entry

Food Name	Calories	Protein	Serving Quantity	Intake Date	Actions
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Add Dietary Entry

Food Name

rice bowl

Calories

500

Protein

40

Serving Quantity

1 bowl

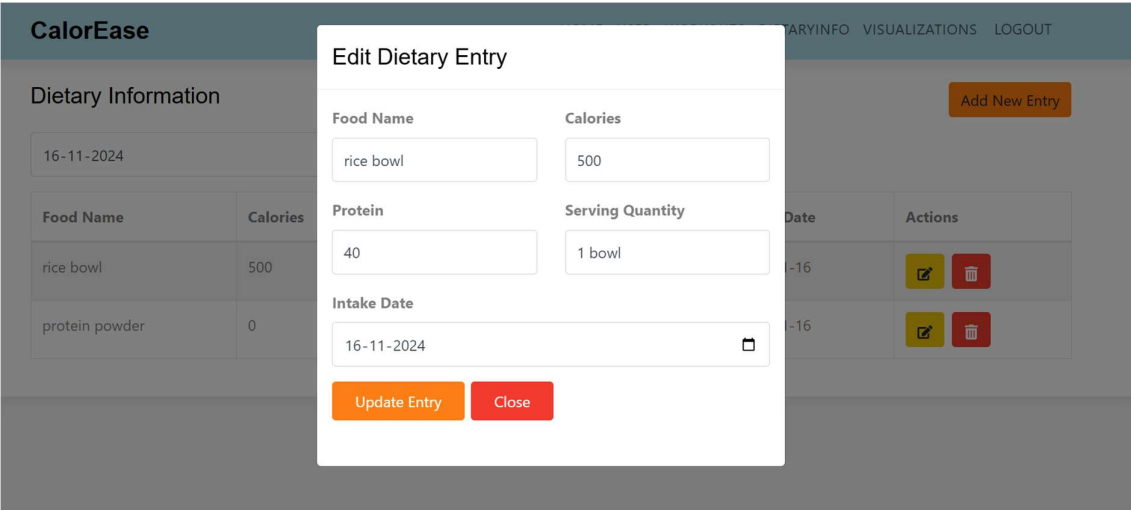
Intake Date

16-11-2024

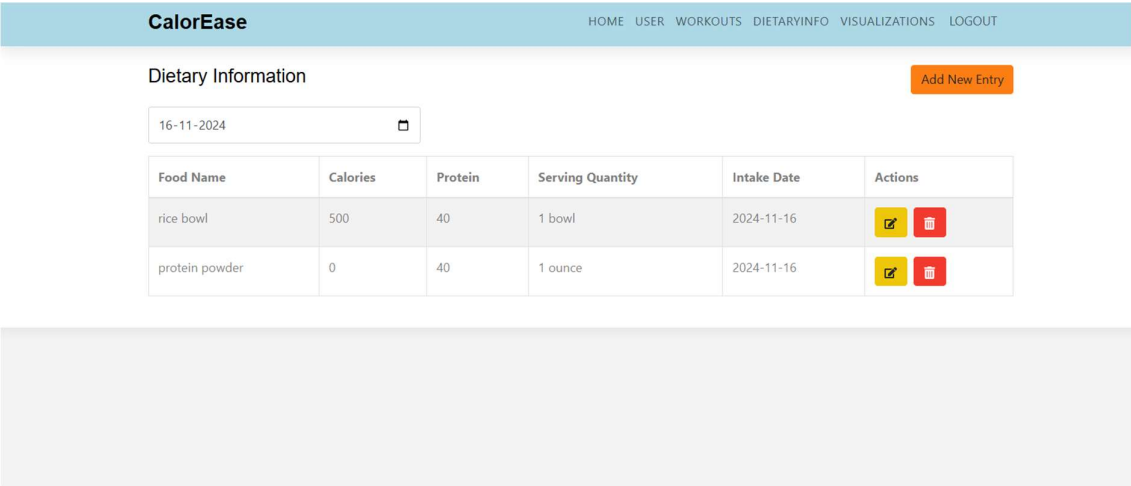
Add Entry

Close

- The table also includes Edit and Delete buttons:
 - Edit Button: Users can edit the details of a meal by clicking on the Edit icon. This allows them to update information such as the meal name, calories, protein, serving quantity, and intake date. Any changes made will be immediately reflected in the table.



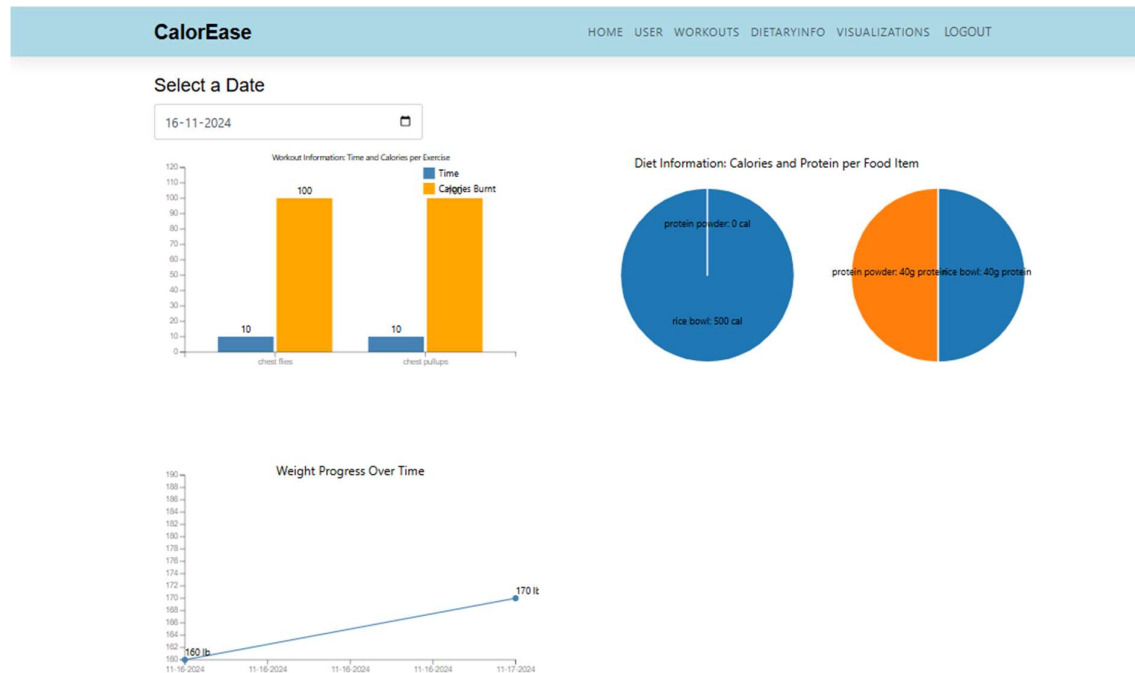
- Delete Button: Users can delete a specific meal entry by clicking the Delete icon. The entry will be removed from the table, ensuring only relevant data is kept.



- **Visualization Page:**

The Visualization Page provides users with graphical representations of their workout and dietary trends. It includes the following charts:

- Bar Graph for Workout Data:
 - This graph displays the calories burned during workouts over time. Users can select a specific date range, and the graph will show the calories burned for each workout on the selected dates.
- Pie Chart for Dietary Data:
 - The pie chart displays the proportion of protein and calories in different food items for the selected date. Users can select a specific date, and the chart will show the breakdown of protein and calorie content for each meal or food item consumed on that date.
- Line Graph for Weight Trends:
 - The line graph shows the user's weight progression over time. Users can view their recorded weights on different dates, with the X-axis representing time and the Y-axis representing weight.



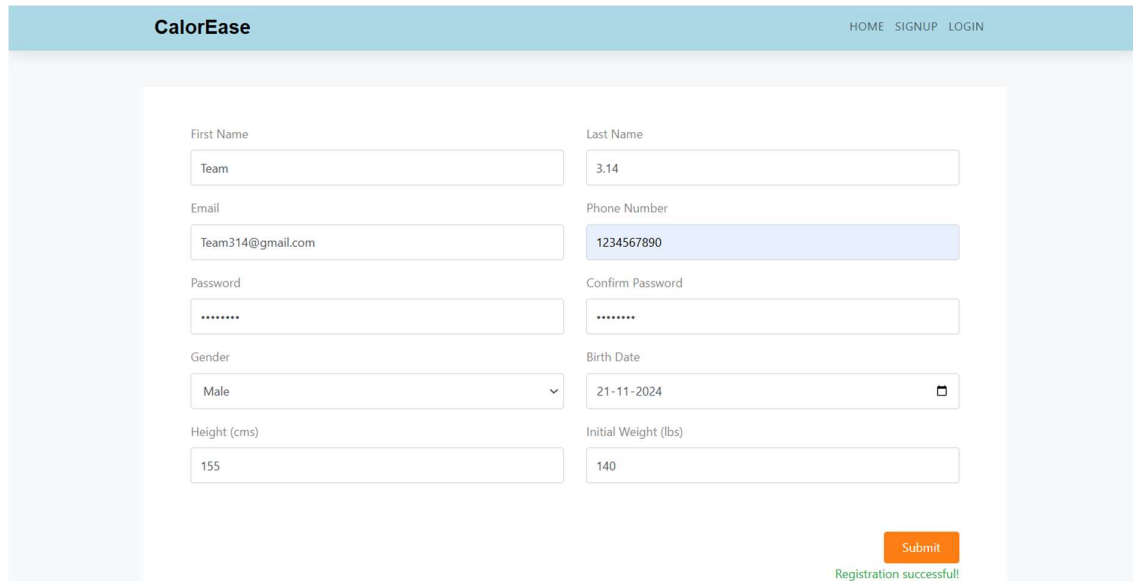
- **Logout**

When the user clicks on the **Logout** button, the system will clear the current session and all associated data. This will log the user out of the application and redirect them back to the **Home Page**, where they will be presented with the **Login** and **Sign-Up** options.

3. Main Scenario Walkthrough: Using Calorease

- **User Signup:**

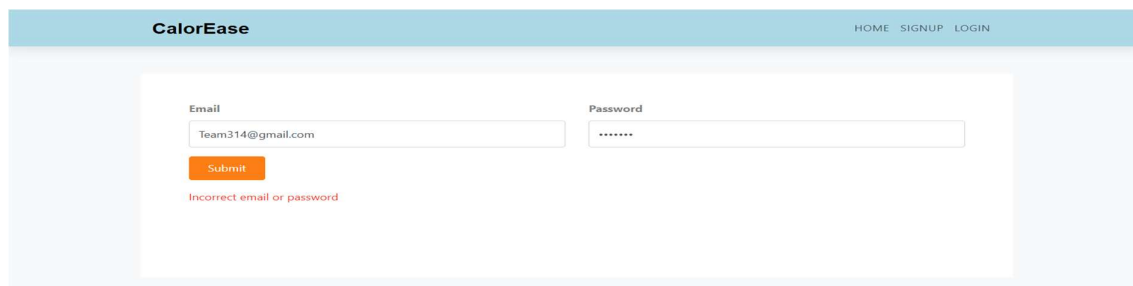
- Action: The user begins by visiting the Signup Page.
- Steps:
 - Enter First Name, Last Name, Email, Phone Number, Password, Confirm Password, Date of Birth, Weight, and Height.
 - Submit the form to create an account.
 - After successful signup, the user is redirected to the Login Page.



The screenshot shows the CalorEase Signup Page. The header has the CalorEase logo on the left and navigation links (HOME, SIGNUP, LOGIN) on the right. The main content area contains a registration form with the following fields: First Name (Team), Last Name (3.14), Email (Team314@gmail.com), Phone Number (1234567890), Password (masked with dots), Confirm Password (masked with dots), Gender (Male), Birth Date (21-11-2024), Height (cms) (155), and Initial Weight (lbs) (140). A blue Submit button is at the bottom right. Below the button, a green message reads "Registration successful!".

- **User Login:**

- Action: After signing up, the user logs into the system.
- Steps:
 - On the Login Page, enter the Email and Password that were used during signup.
 - If the credentials are correct, the user will be logged in and redirected to the Home Page.
 - If the password is incorrect, an error message will be shown.



The screenshot shows the CalorEase Login Page. The header has the CalorEase logo on the left and navigation links (HOME, SIGNUP, LOGIN) on the right. The main content area contains a login form with the following fields: Email (Team314@gmail.com) and Password (masked with dots). A blue Submit button is below the Email field. Below the button, a red error message reads "Incorrect email or password".

CalorEase

HOME SIGNUP LOGIN

Email

Team314@gmail.com

Submit

Password

.....

- **Accessing the Home Page:**
 - Action: After logging in, users are directed to the Home Page.
 - Home Page Features:
 - Users can see options to navigate to User Page, Workout Page, Dietary Page, and Logout.

CalorEase

HOME USER WORKOUTS DIETARYINFO VISUALIZATIONS LOGOUT



USER PAGE
Enables user registration with personal details, email, password, and optional verification for security.



WORKOUT TRACKER
Records the number of calories burned and total minutes spent exercising for each workout session.



DIETARY INFORMATION
Allows users to log daily meals and monitor caloric intake to align with their fitness goals.



- **Managing Profile Information:**
 - Action: On the User Page, users can manage their profile information.
 - Steps:
 - Users can Update Personal Information such as First Name, Date of Birth, and Weight.
 - The Email cannot be changed.

CalorEase

HOME USER WORKOUTS DIETARYINFO VISUALIZATIONS LOGOUT

Workout Information

Add New Workout

Select Date

16-11-2024

Workout	Description	Duration	Calories Burned	Date	Actions
chest flies	Done with support	10	100	2024-11-16	 
chest pullups	Done with support	10	100	2024-11-16	 

- Users can also Change Password or Delete Account.
- The View Weight section displays a table of previous weights, with options to Edit or Delete.

CalorEase

DIETARYINFO · VISUALIZATIONS · LOGOUT

Update Weight

UPDATE USER DATA

First Name

Team

Email

Team314@gmail.com

Gender

Male

Height (cms)

155

Phone Number

1234567890

Birth Date

13-11-2024

Submit

User data updated successfully

Update Weight

Weight updated successfully

Current Weight (lbs)

170

Date

17-11-2024

Cancel

Update Weight

Change Password

Delete User

CalorEase

DIETARYINFO · VISUALIZATIONS · LOGOUT

Update Weight

UPDATE USER DATA

First Name

Team

Email

Team314@gmail.com

Gender

Male

Height (cms)

155

Phone Number

1234567890

Birth Date

13-11-2024

Submit

User data updated successfully

View Weight History

Weight	Date	Actions
170	2024-11-17	
160	2024-11-16	

Cancel

Change Password

Delete User

- **Logging Workouts:**
 - Action: On the Workout Page, users can log their workouts.

CalorEase

HOME · USER · WORKOUTS · DIETARYINFO · VISUALIZATIONS · LOGOUT

Workout Information

Add New Workout

Select Date

dd-mm-yyyy

Workout	Description	Duration	Calories Burned	Date	Actions
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- Steps:
 - Click the Add Workout button to enter the details: Workout Name, Duration, Calories Burned, Date, and Description.

CalorEase

Workout Information

Select Date

16-11-2024

Workout	Description
chest flies	Done with support

Workout Name

chest pullups

Duration (minutes)

10

Calories Burnt

100

Performed On

16-11-2024

Description

Done with support

Add Entry Close

NAVIGATION: HOME USER WORKOUTS DIETARYINFO VISUALIZATIONS LOGOUT

Buttons: Add New Workout, Edit, Delete

- Once added, the workout information will be displayed in a table based on the selected date.

CalorEase

Workout Information

Select Date

16-11-2024

Workout	Description	Duration	Calories Burned	Date	Actions
chest flies	Done with support	10	100	2024-11-16	Edit Delete
chest pullups	Done with support	10	100	2024-11-16	Edit Delete

NAVIGATION: HOME USER WORKOUTS DIETARYINFO VISUALIZATIONS LOGOUT

Buttons: Add New Workout

- Users can Edit or Delete any workout entry as needed.

CalorEase

Workout Information

Select Date

16-11-2024

Workout	Description
chest flies	Done with support
chest pullups	Done with support

Workout Name

chest flies

Duration (minutes)

10

Calories Burnt

100

Performed On

16-11-2024

Description

Done with support

Update Entry Close

NAVIGATION: HOME USER WORKOUTS DIETARYINFO VISUALIZATIONS LOGOUT

Buttons: Add New Workout, Edit, Delete

- **Logging Meals:**
 - Action: On the Dietary Page, users can log meal details.
 - Steps:
 - Click the Add Meal button to enter the Meal Name, Calories, Protein, Serving Quantity, and Intake Date.

CalorEase | DIETARYINFO | VISUALIZATIONS | LOGOUT

Dietary Information

17-11-2024

Food Name | Calories

Intake Date | Actions

Add Dietary Entry

Food Name: rice bowl | Calories: 500

Protein: 40 | Serving Quantity: 1 bowl

Intake Date: 16-11-2024

Add Entry | Close

- The meal data will be displayed in a table.

CalorEase | HOME | USER | WORKOUTS | DIETARYINFO | VISUALIZATIONS | LOGOUT

Dietary Information | Add New Entry

16-11-2024

Food Name	Calories	Protein	Serving Quantity	Intake Date	Actions
rice bowl	500	40	1 bowl	2024-11-16	
protein powder	0	40	1 ounce	2024-11-16	

- Users can Edit or Delete entries as required.

CalorEase | DIETARYINFO | VISUALIZATIONS | LOGOUT

Dietary Information

16-11-2024

Food Name | Calories

Intake Date | Actions

Edit Dietary Entry

Food Name: rice bowl | Calories: 500

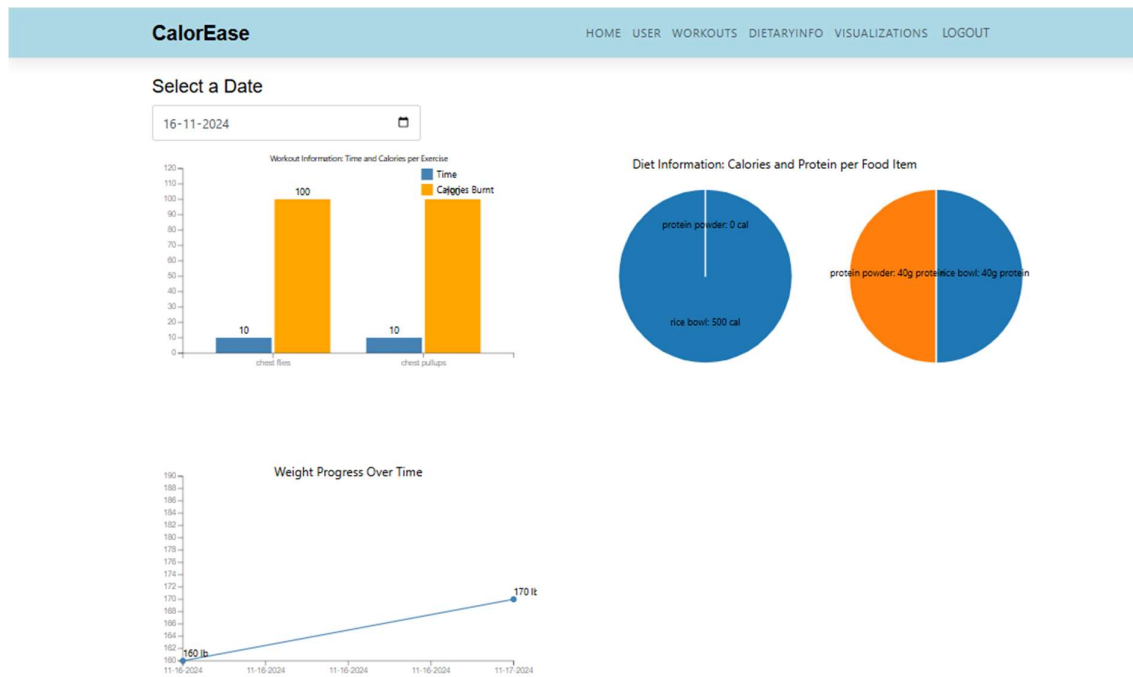
Protein: 40 | Serving Quantity: 1 bowl

Intake Date: 16-11-2024

Update Entry | Close

- **Visualizing Trends:**

- Action: Users can view various visualizations of their workout and dietary data.
- Steps:
 - On the Visualization Page, users will see a Bar Graph showing Calories Burned over time based on workouts.
 - A Pie Chart shows the Protein and Calories breakdown of food items based on a selected date.
 - A Line Graph displays users' weight history over time.



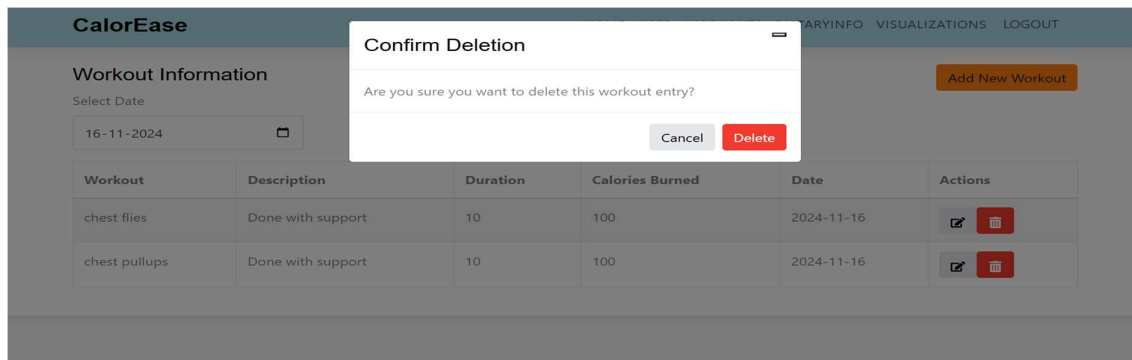
- **Logout:**

- Action: The user can log out of the system at any time.
- Steps:
 - Click on the Logout button.
 - This clears the session and takes the user back to the Home Page, where they can log in again.

4. Additional Scenarios

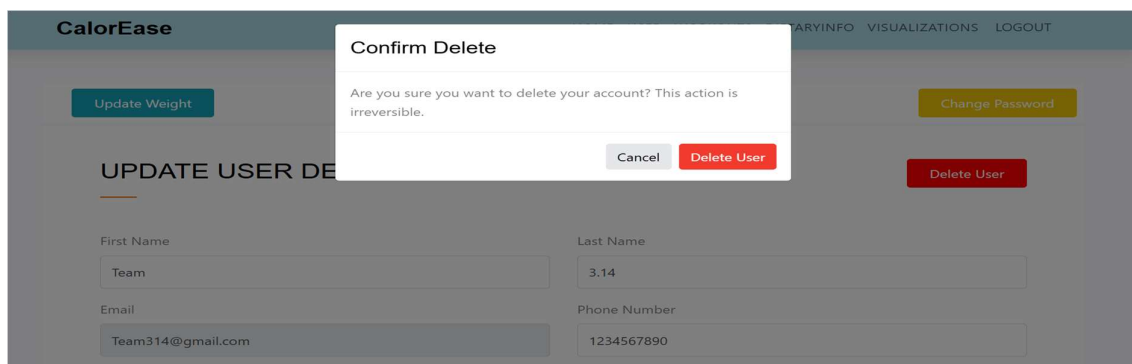
- **User Deletes a Workout Entry:**

- **Action:** The user wants to delete a previously logged workout.
- **Steps:**
 - Navigate to the **Workout Page**.
 - In the table displaying logged workouts, locate the workout entry they wish to delete.
 - Click the **Delete** button associated with that workout entry.
 - A confirmation prompt will appear, asking the user to confirm if they really want to delete the workout.
 - After confirmation, the workout entry will be removed from the table, and the change will be reflected immediately.



- **User Deletes Their Account:**

- **Action:** The user wants to delete their **Calorease** account permanently.
- **Steps:**
 - Navigate to the **User Page**.
 - Click the **Delete Account** button.
 - A confirmation prompt will appear, asking the user to confirm if they are sure about deleting their account. This action is irreversible.
 - After confirming, the system will remove all user data and log the user out, redirecting them to the **Home Page**.



- **Changing User Password**

- **Action:** The user wants to change their **Calorease** account password for security or personal reasons.
- **Steps:**
 - Navigate to the **User Page**.
 - Click the **Change Password** button.
 - The user will be prompted to enter their **current password**.
 - After entering the current password, the user will need to input a **new password** and **confirm** the new password.
 - Click **Save** to update the password.
 - Upon successful password change, a confirmation message will appear indicating that the password has been updated.

The screenshot shows a web application interface for 'Calorease'. A modal titled 'Update Password' is centered on the screen. The modal contains a green success message 'Password updated successfully' at the top. Below it are three input fields: 'Current Password', 'New Password', and 'Confirm Password', each with a masked password (dots). At the bottom of the modal are two buttons: 'Close' and 'Update Password'. The background is a blurred view of the 'UPDATE USER DETAILS' page, which includes fields for 'First Name', 'Team', 'Email' (with the value 'Team314@gmail.com'), 'Gender', and 'Birth Date'. Navigation links like 'PRIMARYINFO', 'VISUALIZATIONS', and 'LOGOUT' are visible at the top right of the background page.